

Abstract:

In today's digital age, social media platforms are plagued by harmful content such as hate speech, bullying, and inappropriate imagery. This not only affects user experience but also poses significant mental health risks. To address this issue, we propose **SafePost**, a social media platform that ensures all user-generated content is harmless and positive. SafePost leverages advanced **Artificial Intelligence (AI)** technologies, including **Natural Language Processing (NLP)** and **Computer Vision**, to automatically detect and filter harmful content before it is published.

Objective:

The primary objective of SafePost is to create a safe and positive social media environment by:

1. Automatically analysing text, images, and videos for harmful content.
2. Providing real-time feedback to users about flagged content.
3. Encouraging positive behaviour through rewards and recognition.

Features:

1. Text Moderation:

- Uses NLP models to detect hate speech, bullying, and offensive language.
- Provides suggestions for revising flagged content.

2. Image/Video Moderation:

- Uses computer vision models to detect inappropriate imagery (e.g., violence, nudity).
- Blocks or blurs flagged images/videos.

3. Real-Time Feedback:

- Notifies users instantly if their post is flagged.
- Explains why the content was flagged and how to fix it.

4. User-Friendly Interface:

- Simple and intuitive design for posting and interacting with content.
- Customizable filters for different sensitivity levels.

5. Admin Dashboard:

- Allows moderators to review flagged content and make final decisions.
- Generates reports on user activity and flagged content.

Technologies Used:

1. Backend:

- Python Flask (not conform) for API development.
- NLP-based text moderation.
- Google Cloud Vision for image/video moderation.

2. Frontend:

- q React.js (not conform) for building the user interface.
- HTML/CSS for styling.

3. Database:

- SQLite or PostgreSQL for storing user data and posts.

4. Deployment:

- Heroku or AWS for backend deployment.
- Netlify or Vercel for frontend deployment.

Workflow:

1. User Uploads Content:

- A user writes a post or uploads an image/video.

2. Content Analysis:

- The text is analysed using NLP models.
- Images/videos are scanned using computer vision models.

3. Decision Making:

- If the content is flagged, the user is notified and asked to revise it.
- If the content is clean, it is published.

4. Admin Review:

- Flagged content is sent to moderators for final review.

Expected Outcomes:

1. A fully functional social media platform that ensures all content is harmless.
2. Reduced exposure to harmful content for users.
3. Improved user engagement through positive reinforcement.

4. Scalable and modular architecture for future enhancements.

Applications:

1. Social Media Platforms: Can be integrated into existing platforms to enhance content moderation.

2. Educational Institutions: Can be used to create safe online spaces for students.

3. Corporate Environments: Can be used for internal communication platforms to ensure professionalism.

Future Scope:

1. Multilingual Support: Extend text moderation to support multiple languages.

2. Advanced AI Models: Train custom models for better accuracy.

3. Mobile App: Develop a mobile version of the platform for wider accessibility.

4. Community Features: Add features like groups, events, and messaging.

Conclusion:

SafePost aims to revolutionize social media by creating a platform that prioritizes user safety and positivity. By leveraging cutting-edge AI technologies, SafePost ensures that harmful content is filtered out in real-time, providing users with a safe and enjoyable online experience. This project has the potential to make a significant impact on how we interact and share content online.

Signature of following member:

Project Members

Project Guide

Project Coordinator

- 1.
- 2.
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